



BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: FarmerBot- An Autonomous Farming Robot
by

Group-2
Section-4

Group Members:

Name	ID
Zyma Zareen Haque	22301781
Mohd Tashwaruddin Safin	21201160
Nazmoon Nahar Aka	22101649
Mustafina Joarder	22101215
Ayesha Akter	23101128

Department of Computer Science and Engineering
BRAC University
5th September, 2025

Abstract

This initiative involves the creation of a self-operating agricultural robot. Its goal is to enhance efficiency, promote sustainability, and simplify usage for farmers. The emphasis was placed on automating processes such as monitoring soil, managing irrigation, and planting seeds. This system reduces manual labor and optimizes resource utilization. The robot is constructed using two Arduino Uno microcontrollers. It features sensors for measuring soil moisture and temperature, a water pump for irrigation, and a servo motor for the seed dispensing mechanism. A four-wheel drive chassis ensures mobility, while obstacle detection allows the robot to navigate safely. Each component adheres to predefined programming rules and collaborates effectively. The robot gauges the soil's moisture content and temperature. It irrigates only when the soil moisture is beneath a set threshold and sows seeds under appropriate soil conditions. The approach consisted of circuit design, sensor integration, code creation, and experimentation in a controlled environment. Irrigation is performed solely when necessary, conserving water in the process. Additionally, it dispenses seeds while avoiding obstacles. This output demonstrates the robot's efficacy in minimizing labor, conserving resources, and saving time. This invention plays a significant role in promoting sustainable agriculture since it employs economical hardware combined with automated decision-making. It is particularly suitable for environments with limited resources and can be utilized by small-scale farmers. The system supports environmental sustainability and contributes to improving global food security.

Keywords

Farming, Arduino, automation, sustainability.

1. Introduction

Conventional farming practices often involve significant manual labor and result in considerable water wastage. Additionally, farming is highly susceptible to climate fluctuations, and many farmers lack access to modern tools, which can lead to inconsistent crop yields. Our innovative robotic solution addresses these challenges by continuously monitoring temperature and soil conditions while automating irrigation processes, thereby minimizing the need for human intervention. The robot features obstacle detection capabilities for safe navigation and the measurement of essential soil parameters, such as moisture levels and temperature. It also automates the irrigation system based on real-time sensor data, has the capability to plant seeds under optimal conditions, and promotes sustainable farming practices while reducing reliance on human labor. This project integrates various technological components, including sensors for obstacle detection, temperature, and moisture measurement, as well as actuators such as DC motors, servo motors, and water pumps, all controlled by Arduino Uno microcontrollers. The implementation demonstrates how computer interfacing can be effectively utilized in robotics to address pressing agricultural challenges.

2. Related Work/ Inspiration

Many innovative farming robots are currently in use, serving as a significant source of inspiration for our project. For instance, Ecorobotix, based in Switzerland, has developed a solar-powered robot designed for precision weeding. Similarly, Naio Technologies from France has created Oz and Dino, robots that assist

with various field operations. Another noteworthy example is Agrobot, which specializes in fruit harvesting by utilizing cameras to identify and pick ripe strawberries. In research settings, projects focused on soil moisture management and seed planting also demonstrate the potential of robotics to alleviate labor-intensive tasks and enhance agricultural efficiency. These advancements have motivated us to design a solution tailored for small-scale farmers.

What distinguishes our project is its emphasis on integration, affordability, and adaptability. Most commercial robots are designed to perform a singular function and often come with a high price tag, making them inaccessible to many small-scale farmers. In contrast, our system integrates soil monitoring, irrigation, and seed planting into one cost-effective robotic solution. By employing open-source hardware, such as Arduino and straightforward sensors, we aim to facilitate use in regions with limited resources or water scarcity. Additionally, the design is structured to support educational initiatives, allowing students to engage with concepts in robotics, the Internet of Things (IoT), and sustainable farming practices. This dual-focused approach not only provides immediate practical benefits but also fosters long-term impact, setting this project apart from existing alternatives.

3. Technical Approach

- **System Architecture:** Autonomous area-coverage farming robot
 1. **Power:** 12V Li-ion battery powering both Arduinos and peripherals (motors, servos, sensors)
 2. **Architecture:** Distributed two-Arduino system
 - a. **Arduino-1:** Navigation & coverage logic
 - b. **Arduino-2:** Soil sensing, watering, and seeding
 3. **Communication:** Simple digital signaling + serial debug; Arduino-1 triggers Arduino-2 via FARM_INPUT pin.

1. Arduino-1 (Navigation / Area Coverage)

Responsibilities:

- Obstacle detection & avoidance (ultrasonic sensor on servo)
- Area coverage following a lawn-mower zigzag pattern
- Motor control via L298N H-bridge
- Virtual position tracking (x, y) and row management
- LCD display of status

2. Arduino-2 (Farming / Actuation)

Responsibilities:

- Soil monitoring (moisture + temperature)
- Automated watering via a water pump
- Automated seeding via servo
- Reporting actions back to Arduino-1 (via Serial)

3. Interaction Between Arduinos

- Arduino-1 triggers or sets FARM_INPUT HIGH at each farming step. Then Arduino-2 reads sensors, acts (water/plant), then resets probes/servos, and finally prints a Serial log of moisture, temperature, watered, and planted

In this robotic system, the two Arduino boards communicate using a straightforward GPIO-based handshake protocol, rather than relying on more complex serial or bus systems. Arduino-1 activates the FARM_INPUT pin by setting it to HIGH whenever it reaches a specific farming step. This action serves as a trigger for Arduino-2 to assess soil conditions, check environmental factors, and execute watering or seeding tasks before returning to its standby mode. This efficient one-bit signaling approach promotes reliable synchronization while minimizing wiring requirements. Additionally, an independent UART (serial communication) is utilized on both Arduinos to transmit debug and status messages to the Serial Monitor. These messages include position updates, obstacle detection logs, and soil condition readings, which facilitate real-time monitoring and verification of the robot's activities during operation.

The block diagram provided below shows how all the components are integrated using 2 Arduino UNO R3 and what purpose each component serves within the whole system.

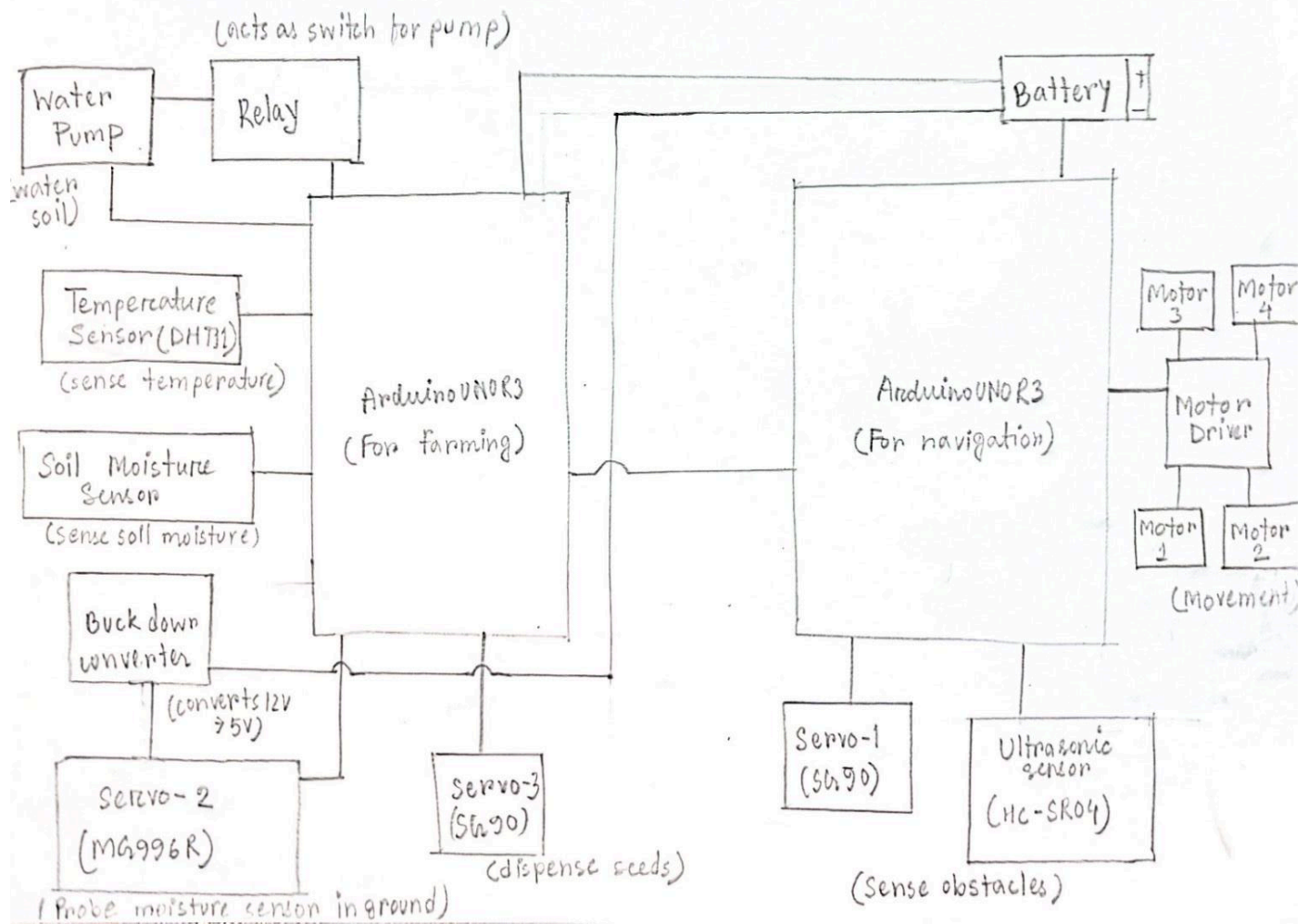


Figure 1: Block diagram of the autonomous farming robot

- **Components Used:**

- **Hardware Components:**

- 1. **Microcontroller:**

- a. **Arduino Uno R3** - Main processing unit responsible for sensor data collection, actuator control, and decision-making algorithms.

- 2. **Sensors:**

- a. **Soil Moisture Sensor** - Measures volumetric water content in soil with resistive sensing technology.
 - b. **Ultrasonic Sensor HC-SR04** - Provides obstacle detection and navigation assistance with a 2-400cm range.
 - c. **DHT11 Temperature Sensor** - Digital temperature measurement for soil thermal monitoring.

- 3. **Actuators:**

- a. **SG90 Servo Motor** - Controls the seed dispenser mechanism with 90-degree rotation.
 - b. **MG996R Servo Motor** - Operates a moisture sensor to test the soil and determine its moisture level.
 - c. **Mini Water Pump** - Provides precise water dispensing for irrigation.
 - d. **DC Gear Motors** - The robotic locomotion system is designed to ensure mobility across a diverse range of terrains.
 - e. **L298N Motor Driver** - Controls DC motor operations and direction.

- 4. **Mechanical Systems:**

- a. **4WD Chassis Kit** - Tracked or wheeled platform providing stability and mobility.
 - b. **Seed Hopper** - Using a funnel to dispense seeds.

- 5. **Power System:**

- a. **12V Li-ion Battery** - Main power source providing portable energy.
 - b. **Buck Converter (12V→5V)** - Voltage regulation for electronic components.
 - c. **JQC3F-05VDC-C Relay** - 5V-controlled switch that lets the low-power signal (from Arduino) control a higher-power load like the pump

- **Software Components:** The system employs the Arduino Integrated Development Environment (IDE) for programming the microcontroller, implementing tailored algorithms for sensor integration, motor control, and automated decision-making processes.

- **Cost Breakdown:**

The autonomous farming robot project exemplifies a commendable level of cost-effectiveness, with a total investment of 5,790 BDT, thereby making it highly accessible for small-scale farmers and educational institutions. The cost structure reflects a strategic allocation of resources that emphasizes essential infrastructure while ensuring affordability through careful selection of components.

The Infrastructure Investment constitutes the largest expense category within the overall budget, amounting to 2,100 BDT. This includes the 4WD Chassis Kit and the 12V Li-ion Battery, which together constitute a significant portion of the total cost. These components form the backbone of the robot's mobility and operational capabilities. The chassis provides a robust platform for field operations, while the high-capacity battery ensures adequate operational duration for extensive field coverage. Additionally, the Buck Converter enhances the power system by supplying stable voltage regulation for sensitive electronic components.

The Processing and Control Systems represent a noteworthy allocation of the budget, with the Arduino Uno R3 microcontroller serving as the central processing unit. This investment offers exceptional value, as the Arduino platform delivers robust processing capabilities, extensive I/O options, and compatibility with a wide array of sensors and actuators, all while adhering to an open-source philosophy that maintains cost efficiency.

The Sensor Technology aspect highlights remarkable cost efficiency within the overall budget, providing comprehensive environmental monitoring capabilities. The Soil Moisture Sensor, Ultrasonic Sensor HC-SR04, and DHT11 Temperature & Humidity Sensor collectively enable precise assessment of soil conditions, obstacle detection, and environmental monitoring. The advantageous cost ratio of these sensors underlines their exceptional value for data acquisition.

The Actuator Systems account for a significant portion of the budget and include a high-performance Servo Motor for demanding operations, an SG90 Servo Motor for precise seed dispensing, and a Mini Water Pump for irrigation purposes. The motor driver ensures vital control of all actuator systems, facilitating precise operation.

Support Components and Accessories constitute an essential part of the total cost, encompassing necessary items such as wires, jumpers, screws, nuts, a breadboard, and various mounting materials. While these components are individually modest in price, they play a crucial role in system integration and ensuring long-term reliability.

The Cost Efficiency Analysis demonstrates outstanding functionality, particularly when considering the three primary functions of soil monitoring, automated irrigation, and seed planting. This approach results in significantly lower costs compared to traditional agricultural automation solutions, which can often reach thousands of dollars.

Overall, the cost structure reflects intelligent resource allocation, with the majority of the investment directed toward durable infrastructure components that provide lasting value. Simultaneously, sophisticated sensing and processing capabilities are attained at minimal cost through careful component

selection. This strategy ensures maximum functionality per dollar invested, while also preserving potential for future upgrades and modifications without requiring substantial additional investment.

- Circuit/Schematic:

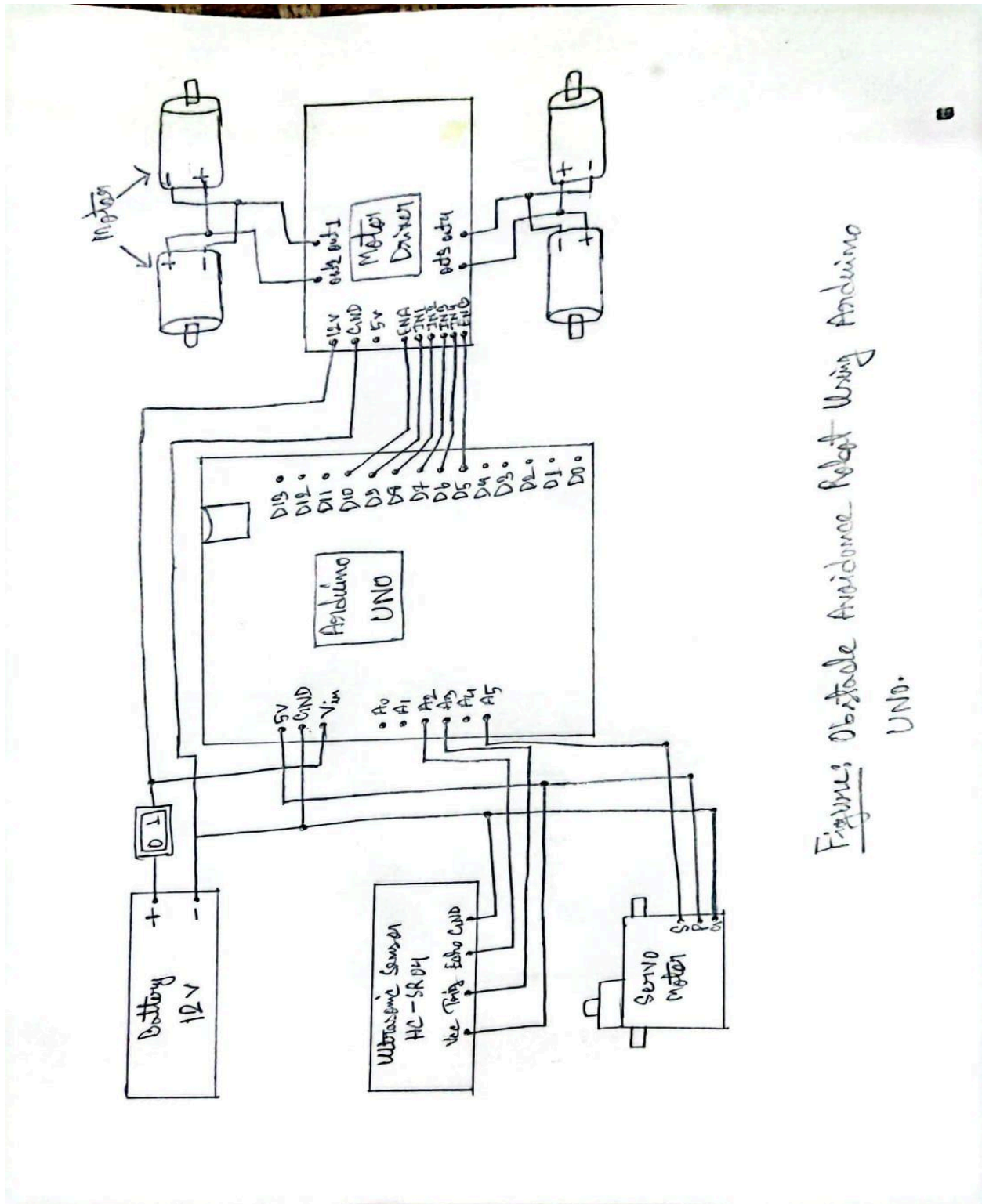


Figure: Obstacle Avoidance Robot Using Arduino Uno.

Figure: Navigation robot block diagram

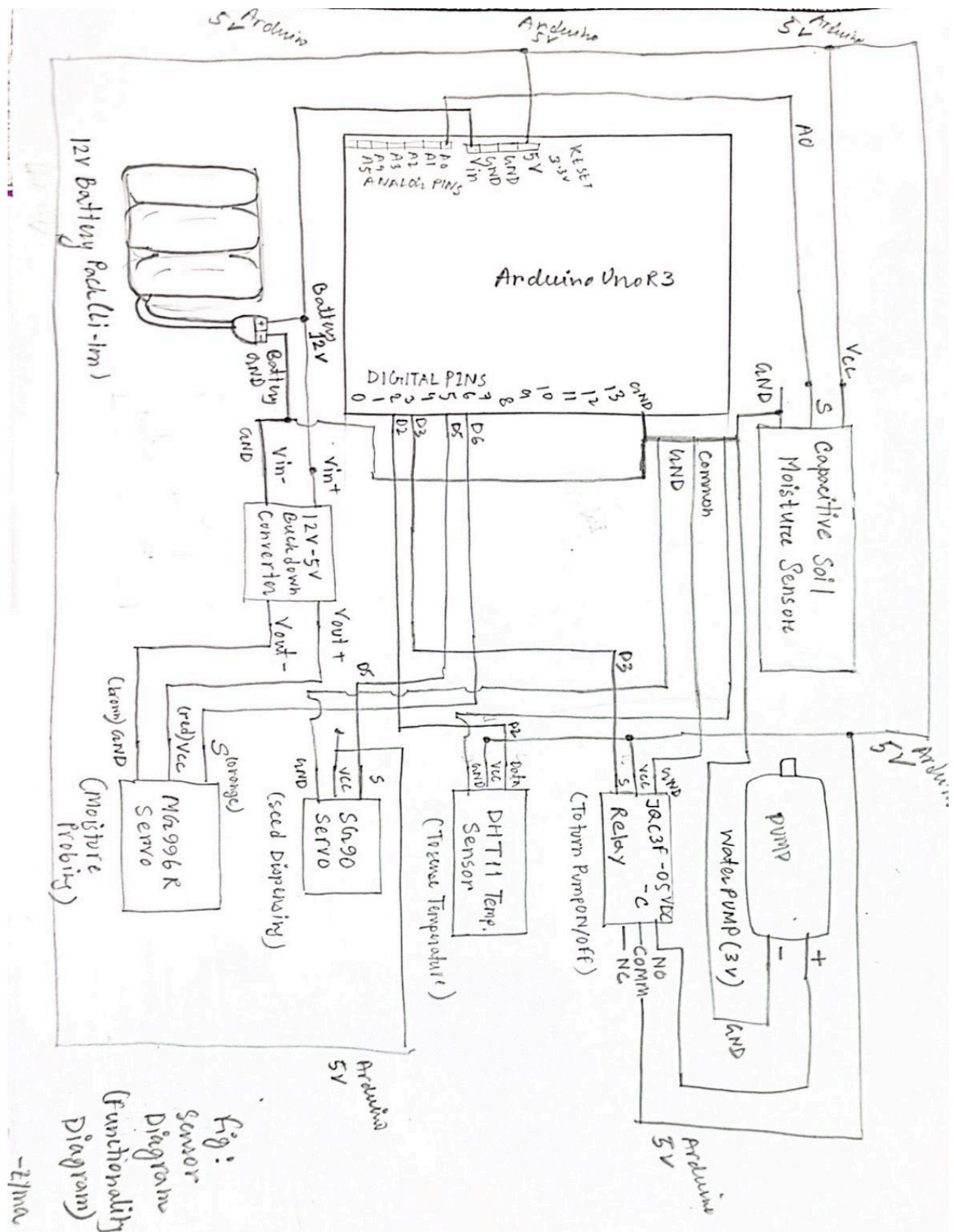


Figure: Sensor diagram

- **Functionality and workflow diagram:**

- **Functionality 1: Soil Condition Monitoring**

- The robot continuously monitors soil environmental conditions to determine optimal planting and irrigation timing:
 - Sensor Deployment - Robot halts at predetermined intervals during field traversal.
 - Data Collection - Soil moisture sensor reads volumetric water content while DHT11 measures temperature.
 - Data Processing - Arduino processes analog/digital signals and evaluates conditions against predefined thresholds.

- **Functionality 2: Automated Irrigation**

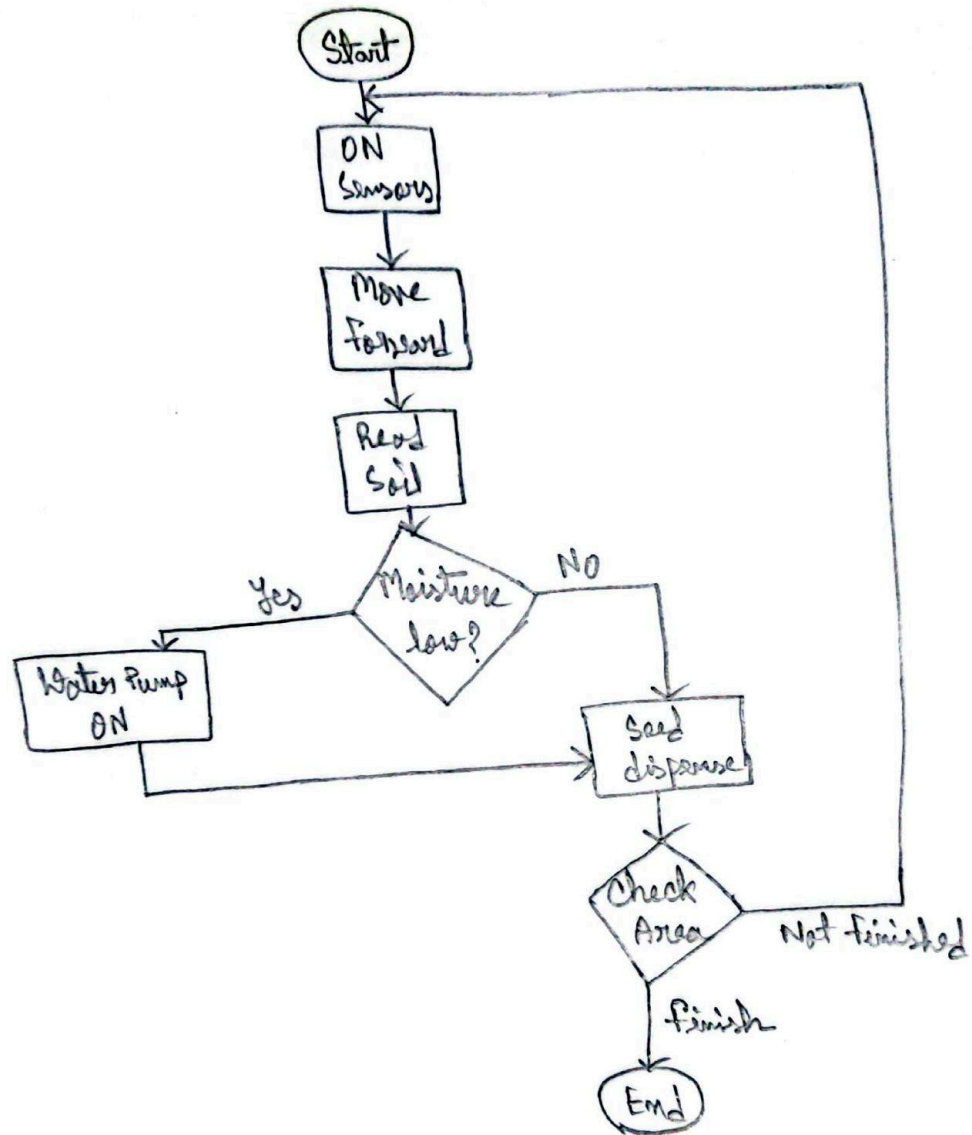
- The system automatically waters the soil when moisture levels fall below critical thresholds:
 - Moisture Detection - Threshold comparison (dry soil < 600 analog value triggers watering).
 - Water Pump Activation - Arduino signals the Relay to activate the mini water pump.
 - Controlled Watering - Pump operates for a fixed duration (3-5 seconds) to prevent overwatering.

- **Functionality 3: Seed Planting Mechanism**

- Seeds are planted only under optimal soil conditions to maximize germination success:
 - Condition Checking - Verifies moisture > 400-600 threshold and temperature within 15°C-35°C range.
 - Seed Dispenser Activation - Servo motor operates hopper mouth when conditions are met.
 - Seed Placement - Seeds fall through the funnel system into prepared soil.
 - Repeat Cycle - Robot advances 20-30cm and repeats the process at the next optimal location.

Workflow Diagram:

Functionality Work-Flow Diagram:



- **Technical Challenges and Solutions**

- **Sensor Accuracy and Calibration**

- **Challenge:** Soil moisture sensors yielded accurate results at times; however, there are instances when their readings may not be reliable.
 - **Resolution:** We have selected high-quality resistive sensors later, equipped with protective coatings to enhance reliability.

- **Power Management**

- **Challenge:** Achieving a balance between battery life and the simultaneous operation of motors, sensors, and water pumps presents a complex challenge.
 - **Resolution:** We have optimized power consumption by employing efficient motor control algorithms and implementing sleep modes during idle periods to extend battery life. Furthermore, we try to use modular systems to minimize the overload of the battery.

- **Motor Control and Navigation**

- **Challenge:** Motors may exhibit non-uniform movement, resulting in deviation from intended paths in the absence of wheel encoders.
 - **Resolution:** We have introduced ultrasonic sensor feedback for course correction and developed straightforward dead-reckoning algorithms to maintain accurate navigation.

- **Design Challenges and Solutions**

- **Mechanical Stability**

- **Challenge:** Designing a stable chassis capable of traversing uneven terrain while accommodating an onboard water tank.
 - **Resolution:** We opted for a tracked chassis design featuring a low center of gravity and a well-distributed weight load to ensure stability.

- **Seed Dispenser Mechanism**

- **Challenge:** Developing a reliable seed dispensing mechanism that prevents jamming and avoids the unintended release of multiple seeds.
 - **Resolution:** We implemented a servo-controlled trapdoor complemented by mechanical stop mechanisms to ensure precise seed delivery.

- **Integration Challenges and Solutions**

- **Component Compatibility**

- **Challenge:** The combination of analog and digital sensors on an Arduino Uno with limited I/O pins complicates integration.
 - **Resolution:** We have used two Arduino Uno R3 and a breadboard to tackle this problem.

4. Sustainability & Impact

- **Sustainability:**

The proposed system enhances environmental sustainability by facilitating precision irrigation and comprehensive soil health monitoring. These features contribute to a reduction in water usage, mitigate nutrient loss, and prevent soil erosion. Additionally, the implementation of electric motors decreases reliance on fossil fuels and helps to minimize air and noise pollution. The use of reusable materials, such as plastics and electronic components, further supports ease of repair and promotes reusability. This approach does not present long-term adverse environmental effects. Moreover, data-driven irrigation practices safeguard groundwater reserves, while the lightweight chassis and controlled movement patterns reduce soil compaction.

- **Impact:**

This system significantly decreases the need for manual labor, an invaluable benefit for small farmers facing labor shortages or related challenges. By incorporating robotic solutions to alleviate heavy labor, the system empowers inclusion among women and the elderly in agricultural activities. Furthermore, optimizing irrigation and planting schedules through data analysis enhances productivity and crop yields, thereby fostering more predictable food supplies and contributing to economic stability. From an industrial perspective, this scalable automation technology can be adapted for precision agriculture, making advanced technology more accessible.

- **Future Work:**

The potential for scalability and increased utility remains promising, with several areas identified for future development:

- Integration of pH and nutrient sensors for more thorough soil analysis and improved decision-making.
- Utilization of artificial intelligence for effective route planning and execution, as well as crop health prediction.
- Integration of solar panels to provide a sustainable power supply in place of conventional batteries.
- Implementation of cloud technology for remote data monitoring and control.

- **Limitations:**

Despite successful implementation, certain limitations have been identified:

- **Hardware Reliability:** The processing power of the Arduino Uno is limited, restricting the integration of machine learning capabilities. Additionally, sensor accuracy may be impacted by fluctuating environmental conditions.
- **Limited Battery Life:** The operation of motors, pumps, and sensors significantly depletes the battery, rendering it unsuitable for long-term deployment.
- **Obstacle Avoidance:** The current HC-SR04 ultrasonic sensor is not equipped to handle complex navigation tasks.
- **Data Privacy:** The data collected is vulnerable unless encrypted and securely transmitted, as it remains local.

- Software Limitations: The robotic system currently relies on a basic "if-else" logical framework, which lacks the adaptability and scalability required for varying conditions.

5. Results & Discussion

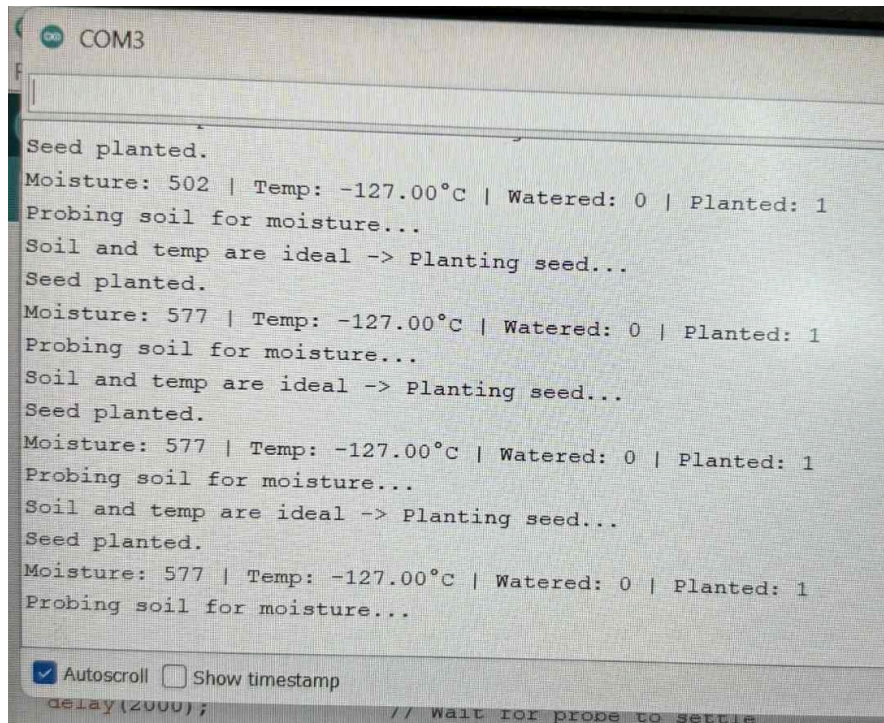
There are two Arduinos used for this robot, one for navigation and one for the actual farming functionalities. The test objectives that we want to achieve from the robot are-

Objective	Measurement Criteria
Verify obstacle avoidance	Success/failure in avoiding obstacles at various distances
Verify area coverage	Percentage of the field covered in a planned zigzag pattern.
Verify soil probing & watering	Soil moisture readings before and after watering
Verify seeding	Seeds planted under suitable conditions

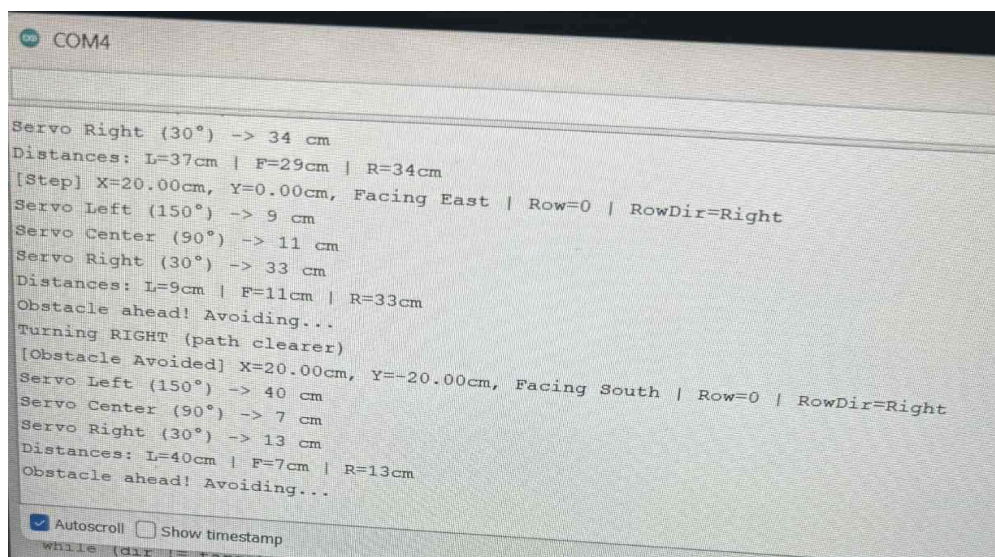
For the farming robot Arduino, the image below shows the serial monitor output values. Here, when the moisture sensor value is below 620, the soil is moist and prepared for seed to be dispersed (Watered=0, Planted=1). But if the moisture level is above 620, the soil is dry and not ideal for seed planting and needs to be watered (Watered=1, Planted=0)

Moisture < 620 → Water Pump ON

Moisture ≥ 620 → Pump OFF



For the Arduino used for navigation, the serial monitor output shows the x and y axis values, direction, and obstacle distances (left, right, and center), and based on that, the robot either moves ahead or backs up. Here, the obstacle threshold was set to 20 cm, so if any obstacle value is below the threshold, the robot will identify that and move accordingly. The serial monitor below shows exactly that.



6. Conclusion

The purpose of this project was to design and implement an autonomous, battery-powered farming robot capable of navigating a field, avoiding obstacles, monitoring soil conditions, and performing essential farming operations such as watering and seeding. Using a dual-Arduino architecture, Arduino-1 handled navigation and area coverage, while Arduino-2 managed soil sensing and actuation, ensuring precise and timely farming actions.

The key achievements of making this robot include reliable obstacle avoidance, automated watering and seeding based on soil moisture and temperature thresholds, and effective field coverage using a zigzag pattern. The system provided real-time feedback through an LCD display and serial monitor, enabling monitoring of position, environmental conditions, and operational status. Synchronization between the two Arduinos allowed seamless coordination of movement and farming actions, demonstrating efficient autonomous operation.

This project highlights the potential of autonomous agricultural robots in promoting sustainable farming. By optimizing water usage, planting seeds under ideal conditions, and reducing manual labor, the robot enhances efficiency while minimizing resource waste. Its modular and scalable design makes it suitable for expansion to larger fields or additional functionalities, offering a practical approach to sustainable, high-efficiency agriculture.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Zyma Zareen Haque	22301781	Hardware design, Circuit building, software design, hardware testing, and software debugging	- Helped write Arduino code for both navigation and farming functionalities of the robot - Helped design the circuit for the robot - Helped implement both the hardware and software design for the robot - Debugged the software code and conducted testing on the hardware
Mohd Tashwaruddin Safin	21201160	Planning the design of the robot, hardware design, Circuit building, software design, hardware testing, and software debugging.	- Assisted in the development of Arduino code for navigation and agricultural applications. - Contributed to the design of the robot's circuitry. - Supported the implementation of both hardware and software systems for the robot. - Engaged in debugging procedures and performed testing on the hardware components.
Nazmoon Nahar Aka	22101649	Wiring internal hardware, constructing the body of the robot, and debugging and testing.	- Contributed to the design of the circuit for the robotic system. - Diagnosed and resolved issues in the code while performing comprehensive testing on the hardware. - Assisted in the implementation of both hardware and software designs for the robotics project.
Mustafina Joarder	22101215	Constructing the body of the robot, debugging, and testing.	- Construct Body - Debugged the code and conducted testing on the hardware
Ayesha Akter	23101128	Constructing the body of the robot, debugging, and testing.	- Construct Body - Debugged the code and conducted testing on the hardware

References

- Muhammad Ansar. (2022, December 13). *Line Follower with Obstacle Avoiding Robot | Arduino and L298 Motor Driver* [Video]. YouTube. <https://www.youtube.com/watch?v=4PQgjjOqJa4>
- ecoRobotix. (n.d.). *ecoRobotix raises USD 14.7 million to speed up series production of its AI-based ultra-high-precision farming solutions*. Venturelab. <https://www.venturelab.swiss/ecoRobotix-raises-USD-147-million-to-speed-up-series-production-of-its-AIbased-ultrahigh-precision-farming-solutions>
- Swiss Food & Nutrition Valley. (2022, February 18). *ecoRobotix: Simplifying farmers' lives with smart solar-powered robots*. <https://swissfoodnutritionvalley.com/ecorobotix-simplifying-farmers-lives-with-smart-solar-powered-robots>
- Modern Farmer. (2020, February 19). *Futuristic farming has arrived with weeding robots*. <https://modernfarmer.com/2020/02/futuristic-farming-has-arrived-with-weeding-robots>
- AgTech Market. (2022, March 4). *Naïo Technologies news update*. <https://www.agtechmarket.net/news/naiolastnews>
- Agrobot. (n.d.). *E-series strawberry harvester*. <https://www.agrobot.com/e-series>